Tugas : Laporan Praktikum Mandiri 6

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```
from google.colab import drive
drive.mount('/content/gdrive')
```

```
Mounted at /content/gdrive
```

```
path = "/content/gdrive/MyDrive/machine learning/praktikum/praktikum6"
```

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv(path + "/data/winequality-red.csv")
df.head()
```

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	alcohol	quality
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	9.8	5
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	9.8	5
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	9.8	6
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	9.4	5

Next steps: Generate code with df New interactive sheet

-Menghubungkan ke google
-Membuat path
-Mengimport modul untuk digunakan
-Menampilkan 5 data pertama

```python
print('Ukuran dataset:', df.shape)
print('\nInfo dataset:')
print(df.info())
print('\nStatistik deskriptif:')
print(df.describe())
print('\nDistribusi nilai kualitas:')
print(df['quality'].value_counts().sort_index())
```

```
Ukuran dataset: (1599, 13)

Info dataset:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1599 entries, 0 to 1598
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   fixed acidity         1599 non-null   float64
 1   volatile acidity      1599 non-null   float64
 2   citric acid           1599 non-null   float64
 3   residual sugar        1599 non-null   float64
 4   chlorides             1599 non-null   float64
 5   free sulfur dioxide   1599 non-null   float64
 6   total sulfur dioxide  1599 non-null   float64
 7   density               1599 non-null   float64
 8   pH                    1599 non-null   float64
 9   sulphates             1599 non-null   float64
 10  alcohol               1599 non-null   float64
 11  quality               1599 non-null   int64  
 12  quality_label         1599 non-null   int64  
dtypes: float64(11), int64(2)
memory usage: 162.5 KB
None

Statistik deskriptif:
       fixed acidity  volatile acidity  citric acid  residual sugar  \
count    1599.000000       1599.000000  1599.000000     1599.000000   
mean        8.319637          0.527821     0.270976        2.538806   
std         1.741096          0.179060     0.194801        1.409928   
min         4.600000          0.120000     0.000000        0.900000   
25%         7.100000          0.390000     0.090000        1.900000   
50%         7.900000          0.520000     0.260000        2.200000   
75%         9.200000          0.640000     0.420000        2.600000   
max        15.900000          1.580000     1.000000       15.500000   

         chlorides  free sulfur dioxide  total sulfur dioxide      density  \
count  1599.000000          1599.000000           1599.000000  1599.000000   
mean      0.087467            15.874922             46.467792     0.996747   
std       0.047065            10.460157             32.895324     0.001887   
min       0.012000             1.000000              6.000000     0.990070   
25%       0.070000             7.000000             22.000000     0.995600   
50%       0.079000            14.000000             38.000000     0.996750   
75%       0.090000            21.000000             62.000000     0.997835   
max       0.611000            72.000000            289.000000     1.003690   
```

-Untuk melihat informasi dasar dataset seperti ukuran, tipe data, dan distribusi nilai target.

```python
df['quality_label'] = df['quality'].apply(lambda x: 1 if x >= 7 else 0)

X = df.drop(columns=['quality', 'quality_label'])
y = df['quality_label']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

print('Distribusi label pada data latih:')
print(pd.Series(y_train).value_counts())
```

```
Distribusi label pada data latih:
quality_label
0    1105
1     174
Name: count, dtype: int64
```

- Mengubah label `quality` menjadi biner: **1 = bagus (≥7)**, **0 = buruk (<7)**.
- Memisahkan fitur dan target.
- Melakukan standarisasi fitur agar setiap kolom memiliki skala yang seimbang.

```python
models = {
    'Linear SVM': SVC(kernel='linear', C=1, class_weight='balanced'),
    'RBF SVM': SVC(kernel='rbf', C=1, gamma='scale', class_weight='balanced')
}

results = {}
for name, model in models.items():
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    acc = accuracy_score(y_test, y_pred)
    results[name] = acc
    print(f'Model: {name}')
    print('Akurasi:', acc)
    print(classification_report(y_test, y_pred, zero_division=0))
    print('-' * 50)
```

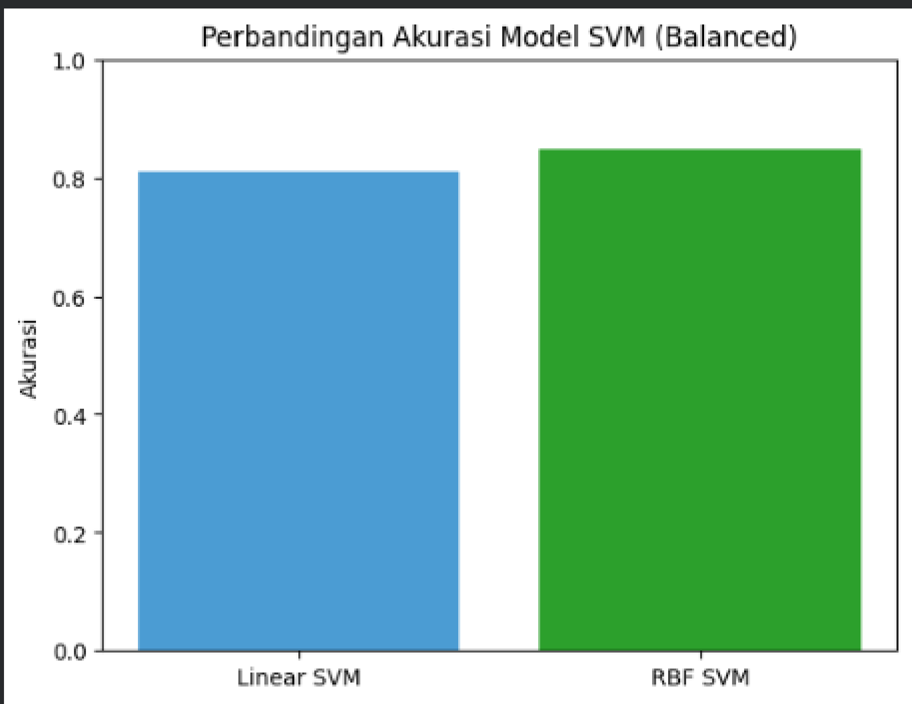
```
Model: Linear SVM
Akurasi: 0.809375
              precision    recall  f1-score   support

           0       0.97      0.81      0.88       277
           1       0.40      0.81      0.53        43

    accuracy                           0.81       320
   macro avg       0.68      0.81      0.71       320
weighted avg       0.89      0.81      0.83       320

--------------------------------------------------
Model: RBF SVM
Akurasi: 0.846875
              precision    recall  f1-score   support

           0       0.96      0.86      0.91       277
           1       0.46      0.77      0.57        43

    accuracy                           0.85       320
   macro avg       0.71      0.81      0.74       320
weighted avg       0.89      0.85      0.86       320

--------------------------------------------------
```

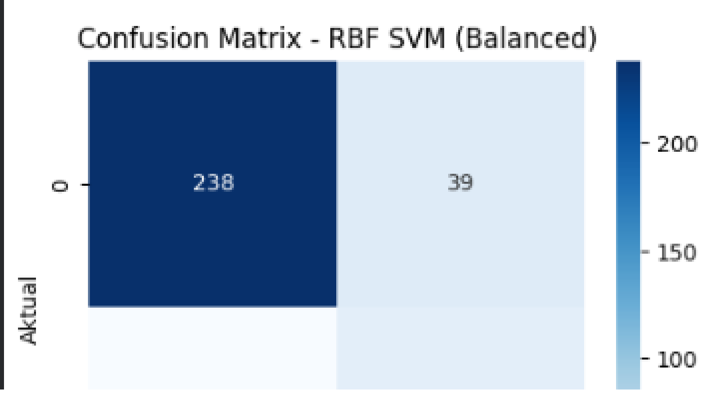
-Model SVM digunakan untuk melakukan klasifikasi dengan dua kernel:

+ Linear
+ RBF (Radial Basis Function)

Parameter `class_weight='balanced'` digunakan untuk mengatasi ketidakseimbangan kelas.

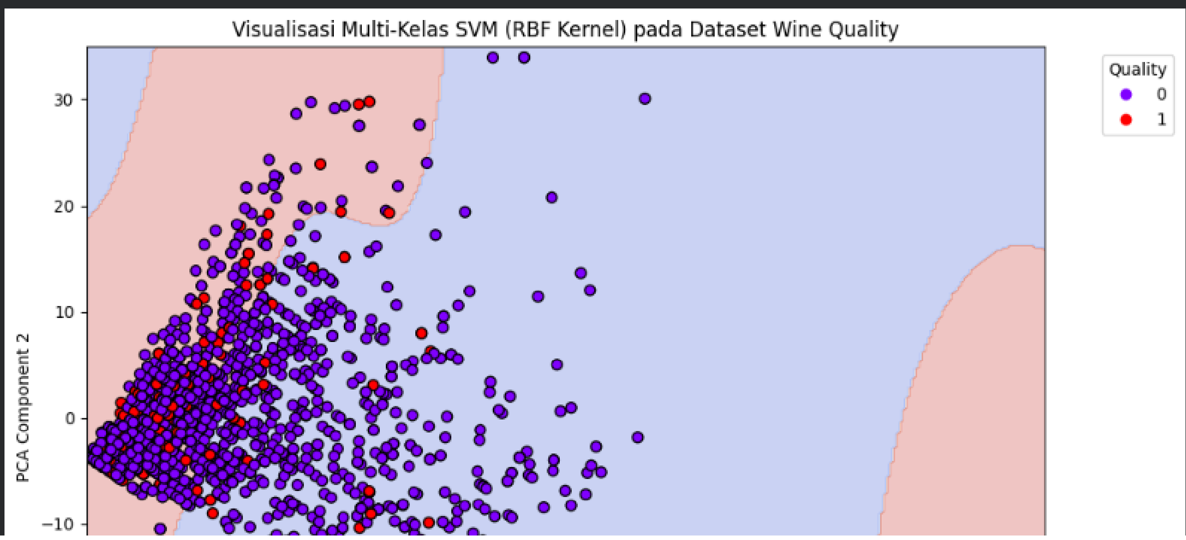
```
plt.bar(results.keys(), results.values(), color=['#4B9CD3','#2CA02C'])
plt.title('Perbandingan Akurasi Model SVM (Balanced)')
plt.ylabel('Akurasi')
plt.ylim(0, 1)
plt.show()

best_model = SVC(kernel='rbf', C=1, gamma='scale', class_weight='balanced')
best_model.fit(X_train, y_train)
y_pred_best = best_model.predict(X_test)

cm = confusion_matrix(y_test, y_pred_best)
plt.figure(figsize=(5,4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues')
plt.title('Confusion Matrix - RBF SVM (Balanced)')
plt.xlabel('Prediksi')
plt.ylabel('Aktual')
plt.show()
```





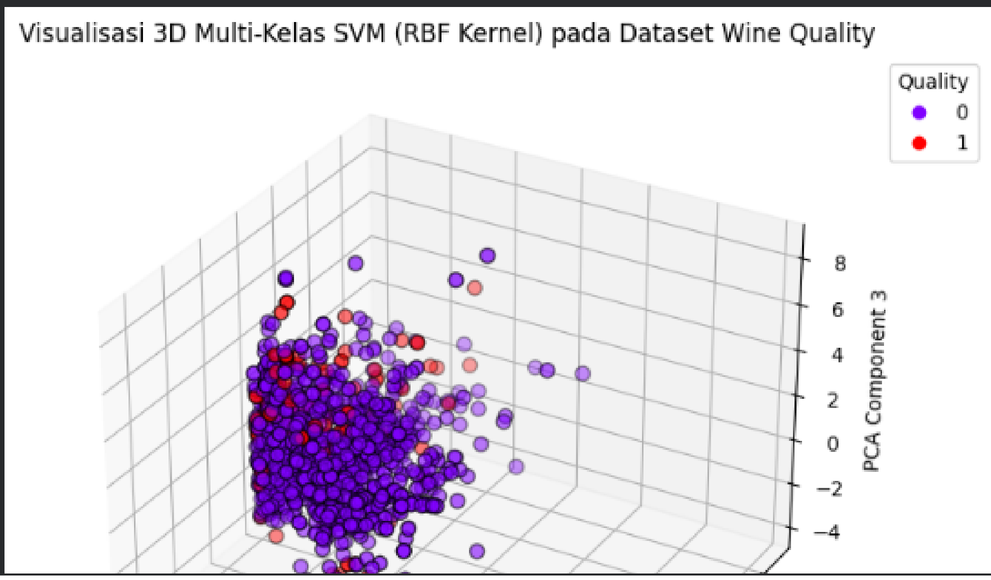
-Hasil akurasi setiap model dibandingkan menggunakan grafik batang. Kemudian, confusion matrix untuk model terbaik divisualisasikan.

```python
# Reduksi dimensi menggunakan PCA ke 2 komponen
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X)

# Latih model SVM dengan kernel RBF (multi-kelas)
svm_vis = SVC(kernel='rbf', gamma='scale', class_weight='balanced')
svm_vis.fit(X_pca, y)

# Buat grid untuk menggambar decision boundary
x_min, x_max = X_pca[:, 0].min() - 1, X_pca[:, 0].max() + 1
y_min, y_max = X_pca[:, 1].min() - 1, X_pca[:, 1].max() + 1
xx, yy = np.meshgrid(np.linspace(x_min, x_max, 300),
                     np.linspace(y_min, y_max, 300))

# Prediksi untuk setiap titik grid
Z = svm_vis.predict(np.c_[xx.ravel(), yy.ravel()])
Z = Z.reshape(xx.shape)

# Plot hasil klasifikasi
plt.figure(figsize=(10, 7))
plt.contourf(xx, yy, Z, alpha=0.3, cmap='coolwarm')
scatter = plt.scatter(X_pca[:, 0], X_pca[:, 1], c=y, cmap='rainbow', edgecolor='k', s=40)
plt.xlabel('PCA Component 1')
plt.ylabel('PCA Component 2')
plt.title('Visualisasi Multi-Kelas SVM (RBF Kernel) pada Dataset Wine Quality')
plt.legend(*scatter.legend_elements(), title="Quality", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```



-Memvisualisasikan data menggunakan model 2D

```python
# Reduksi dimensi ke 3 komponen PCA
pca_3d = PCA(n_components=3)
X_pca_3d = pca_3d.fit_transform(X)

# Latih model SVM untuk data hasil PCA 3D
svm_3d = SVC(kernel='rbf', gamma='scale', class_weight='balanced')
svm_3d.fit(X_pca_3d, y)

# Plot 3D hasil klasifikasi
fig = plt.figure(figsize=(10, 7))
ax = fig.add_subplot(111, projection='3d')

# Titik data dengan warna berdasarkan kualitas wine
scatter = ax.scatter(X_pca_3d[:, 0], X_pca_3d[:, 1], X_pca_3d[:, 2],
                     c=y, cmap='rainbow', edgecolor='k', s=50)

# Label dan judul
ax.set_xlabel('PCA Component 1')
ax.set_ylabel('PCA Component 2')
ax.set_zlabel('PCA Component 3')
ax.set_title('Visualisasi 3D Multi-Kelas SVM (RBF Kernel) pada Dataset Wine Quality')

# Tambahkan legenda di luar plot
legend = ax.legend(*scatter.legend_elements(), title="Quality", bbox_to_anchor=(1.05, 1), loc='upper left')
plt.show()
```



-Memvisualisasikan data menggunakan model 3D

Link GitHub: https://github.com/CeperMail/MachineLearning.git